

Network Hardening — Guided Notes ANSWER KEY

Unit 4.1.3 | CompTIA Network+ N10-009 Obj. 4.2 | N10-008 4.2

For instructor use / distribute after students complete the notes.

Question 1

Network hardening means reducing the _____ surface by disabling unused services, ports, and accounts. List three specific actions that reduce attack surface on a managed switch.

Answer: Attack surface; examples: disable unused switch ports (shutdown), disable Telnet and use SSH only, change default SNMP community strings, disable CDP/LLDP on untrusted ports, enable port security.

Question 2

Port security limits the number of _____ addresses allowed on a switch port. When a violation occurs, the three response modes are _____ (drops frames, no alert), _____ (drops frames, logs alert), and _____ (disables the port).

Answer: MAC addresses; protect; restrict; shutdown (err-disable).

Question 3

802.1X uses three roles: the _____ (client device), the _____ (switch or AP that relays authentication), and the _____ (RADIUS server that validates credentials). What does this prevent that MAC filtering cannot?

Answer: Supplicant; authenticator; authentication server. 802.1X prevents MAC spoofing attacks because it validates identity credentials, not just a hardware address that can be cloned.

Question 4

BPDU Guard protects against rogue switches being added to the network by disabling any access port that receives a _____. This prevents a connected device from participating in _____ elections.

Answer: BPDU (Bridge Protocol Data Unit); Spanning Tree Protocol (STP) root bridge.

Question 5

DHCP snooping builds a binding table of _____ IP-to-MAC-to-port mappings and drops DHCP server messages from _____ ports. Which type of attack does this directly prevent?

Answer: Legitimate/trusted; untrusted. It prevents rogue DHCP server attacks where an unauthorized device hands out malicious IP configurations.

Question 6

Dynamic ARP Inspection (DAI) uses the DHCP snooping binding table to validate ARP packets. It drops ARP replies where the _____ does not match the binding table entry. This defeats _____.

Answer: IP-to-MAC mapping; ARP poisoning / gratuitous ARP attacks.

Question 7

A network baseline documents normal behavior (traffic volumes, protocols, connection patterns) so that _____ can be identified. Name two specific metrics that a baseline should capture.

Answer: Anomalies (deviations from normal); examples: average bandwidth utilization per interface, typical number of active connections per host, expected DNS query volume, normal broadcast rates.

Question 8 — Real World Application

REAL WORLD: A student plugs a cheap consumer router into a classroom Ethernet jack. The router's DHCP server starts handing out 192.168.1.x addresses, overriding the school's DHCP. Name the attack type, the switch feature that would have prevented it, and the command category used to implement it.

Answer: Rogue DHCP server attack. Prevention: DHCP snooping. Implementation: configure the uplink ports to the legitimate DHCP server as trusted (ip dhcp snooping trust) and leave all access ports as untrusted (default), so DHCP Offer/Ack frames from the student's router are silently dropped.